Umberto Tarantino

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he/they

Research interests

Main — Categorical logic, topos theory, duality theory
Others — Intuitionistic logic, model theory, realizability, categorical semantics

Education

Oct 2024 – present — PhD in Theoretical Computer Science · IRIF, Université Paris Cité
Thesis: Classifying topoi and intuitionistic model theory through ultracategories and polyadic spaces
Advisor: Sam van Gool

Sept 2023 – July 2024 — Erasmus+ · Universiteit van Amsterdam

Sept 2022 – July 2024 — MSc in Mathematics · Università degli Studi di Milano
Thesis: A category of arrow algebras for modified realizability
Advisors: Benno van den Berg and Silvio Ghilardi
110/110 cum laude

Sept 2019 – July 2022 — BSc in Mathematics · Università degli Studi di Milano
110/110 cum laude

Publications

2026 — *Toposes with enough points as categories of étale spaces*
Joint work with Sam van Gool and Jérémie Marquès
Journal of Pure and Applied Algebra, 230 (6): 108269

2025 — *A category of arrow algebras for modified realizability*
Theory and Applications of Categories, 44 (4): 132–180

Preprints

2026 — *Conceptual completeness for geometric logic via ultraconvergence spaces*
Joint work with Ivan Di Liberti and Lingyuan Ye

Profunctorial algebras
Joint work with Quentin Aristote

2025 — *Ultracategories via Kan extensions of relative monads*
Joint work with Joshua Wrigley

Teaching

Fall 2026 — Logique · L3 Informatique, *Université Paris Cité*
Exercise classes (in French) · Lecturer: Sylvain Schmitz

Spring 2026 — Introduction à la programmation 2 · L1 Informatique, *Université Paris Cité*
Lab classes (in French) · Lecturer: Daniela Petrişan

Fall 2025 — Programmation 1 · S5 EIDD, *Université Paris Cité*
Lab classes (in French) · Lecturer: Pierre Charbit

Fall 2024 — Programmation 1 · S5 EIDD, *Université Paris Cité*
Lab classes · Lecturer: Pierre Charbit

Spring 2024 Topos theory · MasterMath, *Utrecht University*
Exercise classes · Lecturer: Jaap van Oosten

Community and events

2026 – present Curator of the *ItaCa* website

2025 – present Organizer of the *Realizability reading group* at IRIF

2025 – present Organizer of the *queeRIF* meetings

Dec 2026 Organizer of the *7th ItaCa Workshop*

June 2026 Organizer of the *Ultracategories and Stone Duality Workshop*

2024 – 2025 Organizer of the *Topos theory reading group* at IRIF

Talks

2026 *An arrow-algebraic approach to realizability*
CHoCoLa · *ENS Lyon*

Conceptual completeness for geometric logic via ultraconvergence spaces
XXIX Incontro di Logica AILA 2026 · *Rome*

Conceptual completeness for geometric logic via ultraconvergence spaces
TACL 2026 · *Krakow*

Conceptual completeness for geometric logic via ultraconvergence spaces
Making Categorical Model Theory Accessible · *Masaryk University*

Toposes with enough points as categories of étale spaces
Gothenburg Logic Seminar · *Göteborgs Universitet*

2-dimensional topologies via profunctorial algebras
Stockholm Logic Seminar · *Stockholms Universitet*

Toposes with enough points as categories of étale spaces
7th Workshop on Formal Topology · *Venice*

Toposes with enough points as categories of étale spaces
Séminaires LoCal · *LIPN*

2025 *Toposes with enough points as categories of étale spaces*
ItaCa Workshop 2025 · *Università degli Studi di Milano*

Ultracategories via Kan extensions of relative monads
ItaCa Fest 2025 · *online*

Where do ultracategories come from?
CT2025 · *Masaryk University*

Arrow algebras, algebraic structures for modified realizability
Realizability Workshop · *CIRM, Marseille*

Arrow algebras, algebraic structures for modified realizability
GT Sémantique · *IRIF*

A gentle introduction to categorical realizability
Séminaire des membres non-permanents · *IRIF*

2024 *Triposes and toposes via arrow algebras*
Categorical Logic and Higher Categories · *University of Manchester*

Triposes and toposes through arrow algebras
Dutch Categories And Types Seminar · *Universiteit Leiden*

Awards and scholarships

June 2025 — AILA 3+2 Award
Best Master's Thesis Award of the Associazione Italiana di Logica e sue Applicazioni

Oct 2024 — Contrat International IDEX
Excellence scholarship for international PhD students at Université Paris Cité

Languages

Italian	native
English	fluent, CEFR C1
French	intermediate

Technical skills

Programming	C, Java, F#
Other	LaTeX, Z3, SPASS